

JOSEPH TEDJA NUGRAHA WIBAWA B.Sc.

Full Stack Software Developer

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SUMMARY

Full-Stack Software Developer with 3+ years of experience building optimized, scalable systems. I specialize in translating complex ideas into high-impact applications for companies ranging from early-stage startups to global enterprises. My expertise spans the Education, Real Estate, and Manufacturing sectors, with a core focus on engineering resilient architecture across the entire development lifecycle.

EXPERIENCE

CLOSIO ESTATE, Umalas, Bali

Nov 2025 -

Full Stack Software Developer

Now

Drove ongoing development of a real-estate CRM spanning a PostgreSQL database, NestJS backend microservices, a Next.js web frontend, and a Flutter mobile app, with Docker + GitLab CI pipelines that cut environment-related bugs by ~40% and a TDD workflow that raised feature velocity 50%. Shipped an AI-powered WhatsApp omnichannel assistant that autonomously engages, qualifies, and hands off leads. Cutting first-response time from hours to under 1 minute and automating ~70% of initial lead triage. Built a sandboxed, BullMQ-driven workflow engine that let non-developers author event-triggered rules (e.g. automatic lead routing), eliminating manual assignment for 100% of inbound leads, and an app-wide LLM token-and-cost dashboard that gave full visibility into AI spend and reduced it ~25% through model tuning.

PT MATTEL INDONESIA, Bekasi, West Java

April 2025 -

Software Engineer

Oct 2025

Digitalized Finance and IT processes using ASP.NET, AS400, and Power Automate. Developed REST API integrations, a web inventory audit app, and automated vendor emailing systems. Refactored legacy code, optimized workflows, eliminated manual paper processes, and reduced processing time by 70%.

DIKLATKERJA TEKNO EDUKASI, Bandung, West Java

Jan 2025 -

AI Engineer

April 2025

Developed and optimized Face Recognition and Face Detection systems utilizing DeepFace for the company's viewer detection system. Integrated said systems into the company's frontend Django site. Overall, reduced evaluation time by 84%.

PRESIDENT UNIVERSITY, Bekasi, West Java

Full Stack Web Developer Part-Time

Jul 2023 -

Maintained a Moodle-based PHP web application used by President University with more than 10,000 users. Collaborated in developing an admin site connected to Moodle via Moodle APIs. Handled user creation, class creation, quiz creation, and enrolment, which involves data cleaning and analysis with Python. Reduced enrolment and class creation time by 96%.

Sep 2025

Academic Bureau AI Chatbot Developer Part-Time

Jul 2023 -

Collaborated in developing a web-based AI chatbot integrated with OpenAI API. Built and applied a knowledge-base chatbot context storage scheme. Eliminated emails from students to the academic bureau by 78%.

Dec 2023

SKIN AI, Bekasi, Jawa Barat

Aug 2022 -

Startup Chief Technology Officer

Jul 2023

Designed company app UI/UX as a prototype, and received funding from Setsail. Collaborated in producing a full-stack mobile application utilizing the TensorFlow Lite AI model. Registered an intellectual property right.

DICODING ACADEMY PROJECTS

Aug 2022 -
Apr 2025

Android Development | Model, View, View-Model (MVVM) Architecture

- GitHubUserApp | Kotlin, SQLite, SharedPreferences
- StoryApp | Kotlin, REST Server, Scroll-Pagination, Google Maps API
- StoreApp | Kotlin, Jetpack Compose, Compose Navigation

Backend Development

- Bookstore API | NodeJS, Hapi Framework, AmazonEC2, Postman

Machine Learning Development

- Rock Paper Scissors Image Classification | Python, CNN
- Headline Classification & Beijing Temperature Prediction | Python, NLP, LSTM
- Digit Classification (MNIST Augmented) | Python, TensorFlow, CNN
- Heart Disease Detection | Python, TensorFlow, DecisionTrees, RandomForest, LR, KNN
- Book Recommender System | Python, TensorFlow, TFIDF, Embedding
- Amazon Review Sentiment Analyzer | Python, TensorFlowX, NLP, Docker
- Diabetes Detection | Python, TensorFlowX, Railway, Docker

Multi-Platform Development | Model, View, Provider (MVP) Architecture

- Restaurant App | Dart, Flutter, REST, SharedPreferences, SQLite, Notifications
- Story App | Dart, Flutter, REST, Google Maps API, Scroll-Pagination

PRESIDENT UNIVERSITY PROJECTS

Full-Stack App Development

- Twitter Clone | PHP, MySQL, JQuery, Apache
- Tokopedia Scrapper | PHP, Laravel, PuppeteerJS, MySQL
- Cardi. Patient Management Site | ReactJS, NodeJS, ExpressJS, MySQL, Python.
- Eaterly | Kotlin, Firebase Firestore, Firebase Authentication
- LaundryTallyAI | Kotlin, Jetpack Compose, Flask, Ngrok
- GoTo's Selly Auto Text RAG | Kotlin, Compose, Supabase, PostgreSQL, PgVector, HuggingFace LLM Inference API, VoyageAI Embedding Model API

Aug 2022 -
Aug 2025

EDUCATION

PRESIDENT UNIVERSITY, Bekasi Regency, West Java

Aug 2022 -
Dec 2025

Bachelor in Information Technology, 4.00/4.00

Honors: Attained 100% Jababeka Scholarship

Awards: Acquired 2nd Place in Huawei ICT Competition Cloud Track

Acquired 2nd Place in AyoConnect Hackathon

Acquired 3rd Place in Marketeers Innovation Challenge

Activities: SetSail Business Acceleration

Concentration: Artificial Intelligence

UNIVERSITY OF PADUA, Padova, Italy

Sep 2024 -
Feb 2025

Exchange Student (Fully-Funded IISMA Awardee)

Activities: Research in Malware Family Classification through HSFC.

Courses: Artificial Intelligence, Cybersecurity

LANGUAGES

Indonesian (Native), English (145 DET, C2), Japanese (Elementary), Italian (A1)

PUBLICATIONS

Wibawa, J. ., Rahman, F. F., & Firdaus, M. F. (2023). Utilization of GPT-4 to Improve Education Quality Through Personalized Learning for Generation Z in Indonesia. *IT for Society*, 8(1). <https://doi.org/10.33021/itfs.v8i1.4728>

Wibawa, J. T., Gunawan, R., Suhandi, M., & Setiawan, R. (2023). Development of Marketplace Product Data Mining and Analysis Application as a Reference in Running a Business. *IT for Society*, 8(1). <https://doi.org/10.33021/itfs.v8i1.4724>